

Specialized Multiphysics Training

Statement of Work

NASA Marshall Space Flight Center (“customer”) has engaged Veda Tech LLC (Veda Engineering) to provide specialized workshop in multiphysics. As suitable off-the-shelf training materials are unavailable, Veda Engineering will develop the necessary training materials and tutorials. This invoice presents Veda Engineering's understanding of the required training program and provides the associated costs and timelines for developing and delivering the in-person training at the training location.

The scope of this training includes three distinct modules. Their delivery dates are detailed on the following table. Attendees will engage with highly specialized content spanning the disciplines of nuclear, mechanical, and simulation engineering. To benefit fully from this training, participants should already possess a foundational understanding of these subjects.

Module 1: Steady State Fundamentals Creating “Friendly” Meshes

- Via Abaqus
 - Via generic external CAD software
 - Pros & Cons of Meshes
 - Criticality and source definitions
 - Abaqus and neutronics trade-offs
- Bypassing Monte Carlo neutronics analysis
- Integrating with other transport codes
 - Integrating with first-principles approaches
 - Limitations of non-Monte Carlo neutronics routes

Visualization of Neutronics

V&V for the UM approach

- Catching red flags
- Debugging and benchmarks

Module 2: Steady State Coupled Analysis Coupled Radiation/HT Analysis in Abaqus

- Theory
- Neutron and Gamma heating on components
- Adding Heat Pipes
- Adding Gas Flow
- Adding Heat Exchangers
- Cautions and Red-Flags in Coupled Analysis

Coupled T/M Analysis in Abaqus

- Introduction
- Component level structures analysis
- Radiation induced swelling

Module 3: Transient Analysis Reactor Kinetics Integration

- Startup, shutdown and accidents
- Kinetics vs. Dynamics

Reactor Dynamics

- Introduction to Subroutines
- Startup, shutdown and accidents
- Point-Kinetics and higher order methods

Transients with Heat Removal

Transients with other reactor phenomena